

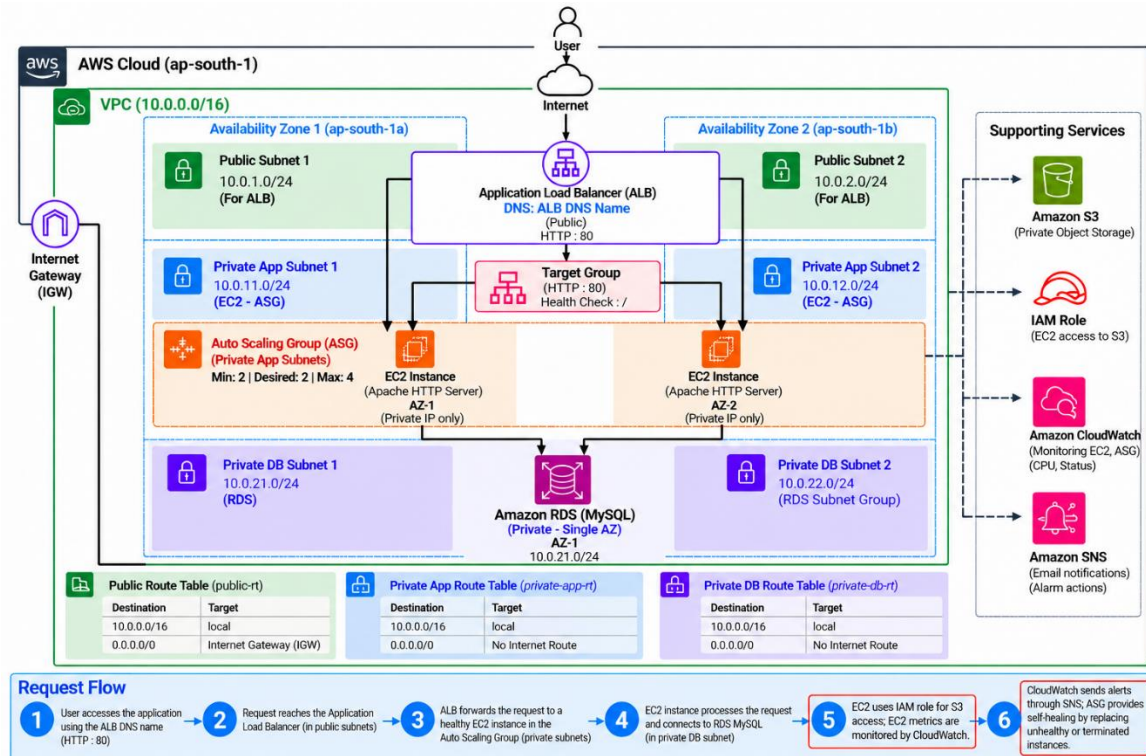
AWS 3-Tier Web Application

AWS Cloud Project | Region: ap-south-1 (Mumbai)

Project Overview

Designed and deployed a **3-tier web application architecture on AWS** using VPC, public/private subnets across two Availability Zones, Application Load Balancer, EC2 Auto Scaling, and Amazon RDS MySQL. Configured secure traffic flow from the **Internet → ALB → private EC2 application servers → private RDS database**. Implemented Auto Scaling self-healing, IAM-based S3 access, and CloudWatch/SNS monitoring.

Architecture:



AWS Services & Key Components:

Service / Component	Purpose
Amazon VPC	Isolated network for the application
Subnets	Public, private application and private database tiers
Internet Gateway	Internet connectivity for public subnets

Route Tables	Control network traffic routing
Security Groups	Control traffic between ALB, EC2 and RDS
Application Load Balancer	Distributes incoming HTTP traffic
Target Group	Routes traffic to healthy EC2 instances
Amazon EC2	Runs Apache application servers
AMI	Reusable image containing the configured application server
Launch Template	Defines EC2 launch configuration
Auto Scaling Group	Maintains capacity and provides self-healing
Amazon RDS MySQL	Private database tier (Single-AZ)
Amazon S3	Private object storage
IAM Role	Provides controlled S3 access to EC2
CloudWatch	Monitoring and CPU alarms
SNS	Email notifications

Security Design:

Internet

↓ HTTP :80

Application Load Balancer

↓ HTTP :80

Private EC2

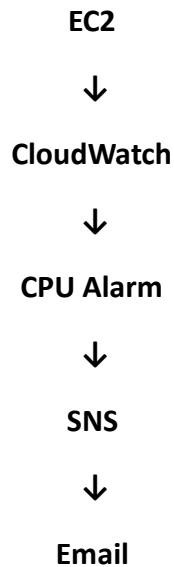
↓ MySQL :3306

Private RDS

- ALB accepts HTTP traffic from the Internet.
- EC2 instances have no public IPs.
- EC2 accepts HTTP traffic only from the ALB security group.
- RDS accepts MySQL traffic only from the application security group.

- S3 Block Public Access is enabled.
- EC2 uses an IAM role instead of long-term access keys.
- RDS encryption is enabled.

Monitoring:



CloudWatch monitors EC2 CPU utilization and sends an email notification through SNS when the configured threshold is exceeded.

Implementation

Networking

- Created VPC 10.0.0.0/16
- Created public, private application, and private database subnets across 2 Availability Zones
- Configured Internet Gateway, route tables, and security groups

Application & Load Balancing

- Configured Amazon Linux EC2 with Apache HTTP Server
- Created AMI and Launch Template
- Configured Application Load Balancer and Target Group
- Deployed EC2 instances using Auto Scaling Group
- **Min: 2 | Desired: 2 | Max: 4**

Database, Storage & Monitoring

- Configured private Amazon RDS MySQL
- Created private S3 bucket with Block Public Access
- Attached IAM role to EC2 for S3 access
- Configured CloudWatch and SNS email notifications

Testing & Results

Test	Result
ALB application access	Passed
Target Group health	2 Healthy
Private EC2 deployment	Verified
ASG across 2 AZs	Verified
ASG self-healing	Passed
RDS private connectivity	Passed
IAM role attached	Verified
CloudWatch alarm & SNS email	Verified

Future Improvements

- **RDS Multi-AZ** — for database high availability and automatic failover.
- **HTTPS using ACM** — secure communication instead of HTTP.
- **Custom domain using Route 53** — instead of accessing the ALB through its DNS name.